

Step 1: Problems and impact:

Problem:

Classrooms in Browning lack storage for student backpacks, forcing students to place them on the floor. Fully loaded backpacks commonly weigh between 10–20 lb, creating safety hazards, property damage, and organizational inefficiencies in classrooms.

Safety	The leading cause of non-fatal injuries for children were falls in 2011 • https://pmc.ncbi.nlm.nih.gov/articles/PMC8404308/ The most common injury event for workers in elementary and secondary schools resulted from falls, slips and trips (67,700 injuries). This measurement was also taken in 2020, during the COVID-19 pandemic, meaning the actual number is likely a much higher than it appears. • https://blogs.cdc.gov/niosh-science-blog/2023/08/10/safety-for-school-workers/ Backpacks on classroom floors meet the definition of environmental tripping hazards identified by national safety organizations.
Property Damage	Bags dragged, stepped on, or crushed under chairs accelerate wear on straps and surfaces, increasing the risk of damage to laptops, textbooks, and other school supplies. Reducing floor contact extends the lifespan of student materials.
Organizational inefficiencies	Teachers lose instructional time reminding students to move bags, and custodial staff must work around obstacles, increasing effort and reducing overall classroom efficiency.

Impact

A desk-mounted backpack hook directly addresses all major issues:

- **Safety:** Keeps walkways clear and reduces tripping hazards
- **Durability:** Prevents backpacks from being crushed or stepped on
- **Organization:** Moves large personal items off the floor and into controlled space
- **Cost efficiency:** Extends the lifespan of backpacks and supplies

- **Scalability:** Can be added without replacing desks or modifying furniture

Step 2: Research

Backpack Weight and Load Considerations:

- Recommended weight limits for backpacks are approximately 10-15% of body weight; however, many students exceed this limit and are unaware of the guidelines. This generates significant dynamic loads when placed quickly or bumped.
 - <https://pmc.ncbi.nlm.nih.gov/articles/PMC3963637/>
- Any hook must safely support at least 20 lbs while resisting deformation.

Desk Geometry:

- Desks in the engineering lab are approximately 1.75 inches in thickness
- A solution must fit multiple desk types without tools or permanent modification.

Average Backpack Cost:

- On average backpacks are about 50\$. A backpack holder will be able to significantly extend the lifetime of the backpack.
 - <https://initibag.com/average-price-for-a-backpack/#:~:text=You'll%20notice%20the%20average%20price%20for%20a,on%20brand%2C%20features%2C%20and%20where%20you%20shop.>

Existing solutions:

Product Name	Manufacturer / Seller	Installation Method	Desk Compatibility	Material	Key Limitations	Link
Backpack Hook for Allied Tables	Demco	Screw-mounted	Specific desk models only	Steel	Requires permanent installation, desk-specific	https://www.demco.com/bookbag-hook-for-allied-tables
Interchange Backpack Hook	Smith System	Screw-mounted	Open-front desks only	Metal	Not universal, requires tools	https://www.acpdirect.com/Smith-System-67638-Inter

						change-Backpack-Hook-for-Open-Front-Desks-P9856.aspx
Student Desk Pack Hook	MooreCo	Screw-mounted	Select MooreCo desks	Steel	Desk-specific, permanent	https://www.moorecoinc.com/product/student-desk-pack-hook/
Classroom Select Backpack Hook (Titanium)	School Specialty	Screw-mounted	Limited desk types	Metal	Permanent install, thin profile	https://www.schoolspecialty.com/classroom-select-backpack-hook-titanium-2038890
WB Mfg. Backpack Hook	WB Manufacturing	Screw-mounted	UNE-T & ELO desks only	Steel	Extremely limited compatibility	https://www.schoolsin.com/wisbph.html
Aluminum Desk Bag Hook	Generic (Walmart)	Slip-on	Variable, unstable	Aluminum	Slips on smooth desks, low load margin	https://www.walmart.com/ip/17832400858
Sturdy Metal Desk Bag Hook	Generic (Walmart)	Slip-on	Variable	Aluminum	Poor fit consistency, no load rating	https://www.walmart.com/ip/18125956076
NeoMove Backpack Hook	Classroom Select	Clip-on	Thin desk edges only	Plastic	Thin plastic, prone to snapping	https://www.schoolspecialty.com/classroom-select-neo-move-backpack-hook-gray-20513

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Limitations of existing solutions (see table above):

- Permanent installation using screws is common, which many schools avoid due to furniture damage and maintenance costs.
- Desk-specific designs limit compatibility and scalability.
- Slip-on hooks often lack sufficient friction or structural thickness, causing slipping or breakage under realistic backpack loads.
- Thin plastic profiles prioritize low manufacturing cost over durability and load capacity.

While these products demonstrate that the problem is recognized, none provide a universal, non-damaging, high-strength solution suitable for widespread classroom deployment.

Identified Design Constraints:

- Must not damage desks or require permanent installation
- Must install and remove without tools
- Must support ≥ 20 lbs under dynamic loading
- Must accommodate desk thicknesses between 1–2 inches
- Must be manufacturable as a single 3D-printed part
- Must be compact and unobtrusive in a classroom setting

These constraints guided all subsequent design decisions.

Step 3: Possible Solutions

Concept 1: Simple Over-the-Desk Hook

A rigid, single-piece hook that slides over the desk edge and hangs downward to support a backpack.

Advantages

- Minimal complexity
- No moving parts
- Easy to model and 3D print
- Tool-free installation

Limitations

- Requires sufficient cross-sectional thickness to avoid snapping
- Fit must balance grip and ease of removal

Concept 2: Over-the-Desk Hook with Padding

A variation of Concept 1 with added padding or textured contact surfaces at desk contact points.

Advantages

- Reduces slipping on smooth desk surfaces
- Protects desk edges from wear
- Same advantages as from concept 1

Limitations

- Slightly increased material and manufacturing complexity
- Padding must remain durable under repeated use

Concept 3: Clamp-Style Hook

A hook that mechanically clamps onto the desk edge using adjustable components.

Advantages

- Very secure attachment
Minimal movement under load

Limitations

- Increased part count and failure points
- More difficult to manufacture as a single printed piece
- Reduced ease of use for students

Step 4: Select a solution

Selected solution: single-piece over-the-desk backpack hook with a thickened structural profile and integrated contact padding at the desk interface (concept 2).

The hook slides over the desk edge and supports the backpack below the desktop surface without requiring screws, adhesives, or tools.

Rationale for Selection:

- This design was selected because it best satisfies the identified constraints while addressing key shortcomings observed in existing solutions:
- Improved Stability:
 - The padded contact surface increases friction between the hook and the desk, reducing the likelihood of slipping on smooth or laminated desk surfaces.
- Desk Protection:
 - Padding prevents surface wear or indentation, addressing concerns schools have about furniture damage.
- Structural Reliability:
 - The load-bearing portion of the hook remains a single solid piece with increased cross-sectional thickness, ensuring sufficient strength under dynamic backpack loads.
- Low Complexity:
 - The padding does not introduce moving parts or additional failure modes and does not affect installation or removal.

Manufacturability:

The hook body is designed to be 3D printed as a single part. Padding can be implemented as:

- an integrated flexible print region, or
- a thin bonded layer applied post-printing without mechanical fasteners.

Both approaches preserve manufacturability while improving performance.

Example of Product:



Step 5: Test and evaluate

In the second iteration of the design, the hook was repositioned to the **inside of the desk** to reduce obstruction for students. Additionally, several edges and corners were extruded or smoothed to minimize sharp angles and improve overall structural integrity.

The design was evaluated conceptually against the original criteria:

1. **Safety and Obstruction:** Placing the hook on the inside reduces interference with student legroom and movement while still providing a clear space for backpacks.
2. **Structural Reliability:** Extruding corners and smoothing edges helps distribute stress more evenly, reducing potential weak points where the hook could fail under typical backpack loads.
3. **Desk Compatibility:** The padded contact surfaces remain intact, ensuring the hook can grip desks securely without causing scratches or damage.
4. **User Interaction:** The inside positioning makes it less intrusive visually and physically, while maintaining easy, tool-free placement and removal.

Evaluation Summary:

The second iteration improves usability by reducing obstruction and potential failure points while retaining the key advantages of the original over-the-desk hook concept. Further iterations could refine edge geometry or padding thickness to optimize stability and comfort.